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in order to be awarded the Degree of
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System Design

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Abstract

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Preface

"I solemnly swear that I'm up to good" (incorrectly inspired by J.K. Rowling)

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List of Abbreviations

AA	Authorization Authority
AT	Authorization Ticket
C-ITS	Cooperative Intelligent Transport Systems
C-V2X	Cellular Vehicle-to-Everything
CAM	Cooperative Awareness Message
CRL	Certificate Revocation List
DAA	Direct Anonymous Attestation
DENM	Decentralized Environmental Notification Message
DSRC	Dedicated Short-Range Communications
EA	Enrolment Authority
ETSI	European Telecommunications Standards Institute
HB	Heartbeat
ITS	Intelligent Transport System
ITS-G5	ITS communications standard for the 5.9 GHz band
ITS-S	ITS Station
MDS	Misbehaviour Detection System
OBU	On-Board Unit
OMNeT++	Objective Modular Network Testbed in C++
OSR	Order for Self-Revocation
PKI	Public Key Infrastructure
PRL	Pending Revocation List
RA	Revocation Authority
RCA	Root Certificate Authority
REWIRE	Revocation Without Resolution
RSU	Roadside Unit
SCMS	Security Credential Management System
SHB	Single-Hop Broadcast
SUMO	Simulation of Urban MObility
TC	Trusted Component
TEE	Trusted Execution Environment
TLS	Transport Layer Security
TPM	Trusted Platform Module
V2X	Vehicle-to-Everything

Chapter 1

Introduction

1.1 Motivation

Vehicle-to-Everything (V2X) communication is being deployed specifically because it is safety-relevant: Cooperative Awareness Messages (CAM) and Decentralized Environmental Notification Messages (DENM) let vehicles warn each other of hazards fast enough to influence automated braking and collision avoidance decisions, as detailed in Chapter 2. This safety dependency is exactly what makes the underlying Public Key Infrastructure (PKI) a hard requirement rather than a defensive add-on: a vehicle’s automated systems cannot act on a warning message without some assurance that the message is authentic, and they cannot verify authenticity without a functioning certificate infrastructure sitting underneath every safety message on the road.

Revocation is the part of that infrastructure most directly exposed to an adversary, and the part most often evaluated as an afterthought. Once a vehicle is known to be compromised, misbehaving, or otherwise untrustworthy, the certificate infrastructure’s only tool for removing it from the network is revocation — and until revocation takes effect, the vehicle’s messages remain cryptographically indistinguishable from a legitimate participant’s. The published literature on V2X revocation, as surveyed in Chapter 2, has overwhelmingly measured revocation schemes on a single axis: how fast do they work under ordinary network conditions? That is a necessary question, but it is not the question an attacker asks. An attacker asks a narrower one: given that I know exactly how this particular scheme decides I am revoked, what is the latest possible moment at which that becomes true for me, and can I push it later still? Answering that question requires an attacker who understands each scheme’s enforcement mechanism specifically, not one who is merely a source of background packet loss. This distinction — generic network unreliability versus a deliberate, scheme-aware attacker — is the motivation for this thesis, and it is the axis along which prior work on this exact codebase, discussed next, does not yet extend.

1.2 Problem Statement

Dehaes [6] and Orban [15] implemented all three revocation schemes under study here — active, passive, and self-revocation — side by side on the Artery/SUMO/OMNeT++ stack this thesis continues to use, and produced the only existing comparative evaluation of the three on a common testbed. Their evaluation, however, is restricted to benign network conditions: probabilistic packet loss and delay, applied uniformly at the RA and radio layers to approximate ordinary network unreliability, with no vehicle ever behaving adversarially with respect to the revocation mechanism itself. Chapter 2 shows that this is not a gap unique to their thesis — even Scopelliti et al.’s own adversarial evaluation of self-revocation [16] tests only that one scheme in isolation against a revocation-aware attacker, and Angelogianni et al.’s comparative work [2] compares architectures under normal operation, not revocation schemes under attack. No source identified in this literature review evaluates active, passive, and self-revocation together under a common, deliberately adversarial condition.

This thesis addresses that gap directly. It asks how each of the three revocation schemes degrades when the packet loss or delay a vehicle experiences is no longer a random network condition but a strategy chosen by an attacker who specifically wants to evade or delay its own revocation, and — the comparative question the existing literature cannot answer — which scheme remains most robust once that shift is made. Speed under benign conditions and robustness under attack are not the same property, and Dehaes and Orban’s own result that active and self-revocation achieve near-identical revocation times under benign conditions is, if anything, a reason this comparison matters: two schemes that look equivalent when nothing is trying to break them may not look equivalent once something is.

1.3 Research Questions

RQ1. Under attacks that target each scheme’s own enforcement mechanism — rather than generic network unreliability — how do active, passive, and self-revocation compare in effective revocation time, message overhead, proportion of vehicles correctly revoked within a fixed time bound, and false-negative exposure window?

RQ1 is answerable directly by the methodology already defined in Chapter 3: the included attack catalog (coverage-gap, selective forwarding, CRL flooding, re-enrolment bypass, targeted heartbeat suppression, and trusted-time manipulation, with jamming and heartbeat replay included conditionally) was selected specifically to produce discriminative, measurable outcomes across at least two of the three schemes (Section 3.2.1), and the metric set — effective revocation time and message overhead inherited from Dehaes and Orban, extended by proportion of vehicles revoked within a time bound and false-negative rate — is already fixed for exactly this purpose.

RQ2. Which structural properties of a scheme’s enforcement mechanism — push-based broadcast dependency versus pull-based local enforcement, explicit peer-side verification versus implicit desynchronisation, and central-authority trust boundary — explain differences in robustness observed under RQ1, as opposed to differences in raw revocation speed under benign conditions?

RQ2 follows from a pattern already visible in the attack-by-attack analysis underlying Chapter 3: passive revocation’s pull-based, expiry-driven enforcement makes it structurally unreachable by broadcast-disruption attacks that directly threaten active and self-revocation, not because it is a faster or better-engineered scheme in the abstract, but because its enforcement point sits somewhere those attacks cannot touch. RQ2 asks the thesis to state that architectural explanation explicitly, rather than leave the comparative ranking in RQ1 as an unexplained numerical result.

RQ3. For each attack found to meaningfully degrade a scheme, what detection signature or mitigation is available, and does deploying it change the comparative robustness ranking established by RQ1 — for instance, by imposing an overhead or latency cost on a scheme that would otherwise appear strictly more robust?

RQ3 operationalises this thesis’s commitment to an attack-and-defense analysis rather than a purely destructive evaluation. It is treated throughout this thesis as a discussion-level question, answered in the Check phase (Chapters 6 and 7) once experimental results from the Do phase (Chapter 4) are available, rather than as a pre-registered part of the attack selection methodology in Chapter 3, which is scoped to attack implementation and measurement, not countermeasure design.

1.4 Contributions

This thesis makes three contributions, each stated here relative to the specific prior work it extends.

A comparative adversarial evaluation of all three revocation schemes on a common testbed. Where Dehaes and Orban compared the three schemes without an attacker, and Scopelliti et al. evaluated an attacker against only one scheme, this thesis holds both constant simultaneously: the same simulation stack, traffic model, and topology as the benign-only comparison, subjected to the same attacker capability across all three schemes in turn.

Attack implementations grounded in the published protocol design and, where the design itself is silent, in direct inspection of this specific codebase — not in generic network degradation. Each attack in the catalog defined in Chapter 3 is grounded in one of two ways: either in a property of a scheme’s actual design, independently confirmed against the published literature (for example, self-revocation’s exclusive reliance on timestamp desynchronisation for peer-side enforcement, and the heartbeat format’s absence of replay protection, both verifiable in Scopelliti et al.’s formalisation of the protocol [16]), or in a security control that no cited standard treats as optional but that Dehaes and Orban’s implementation omits (for example, rate-limiting and cryptographic binding on enrolment requests). Findings of the second kind are corrected before their corresponding experiment is run, so that the comparison measures each scheme’s real enforcement logic rather than an artifact of an unfinished simulator — they are not presented as vulnerabilities discovered through this thesis’s own analysis. This distinguishes the thesis’s attacker from a probabilistic packet-loss model: every included attack targets a specific, named, and sourced property of how a specific scheme decides a vehicle is revoked.

An attack-and-defense reading of the results. Consistent with RQ3, this thesis does not stop at demonstrating that an attack degrades a scheme. Where an attack succeeds, the thesis analyses what a detector or mitigation for that specific attack would need to observe, and — where the mitigation itself has a cost — whether adopting it changes which scheme is judged most robust overall.

1.5 Non-Goals

The following are explicitly out of scope, stated here so that their absence later in the thesis is read as a deliberate scoping decision, not an omission:

- **Proposing a new revocation scheme.** This thesis evaluates the robustness of the three existing schemes as implemented; it does not design a fourth scheme, a hybrid scheme, or a modification intended to outperform the three under test. Where a mitigation is discussed under RQ3, it is discussed as a modification to an existing scheme’s enforcement logic, not as a new scheme.
- **Full ETSI TS 102 941 compliance.** The codebase inherited from Dehaes and Orban makes deliberate simplifications relative to a full ETSI-compliant deployment — a single RSU, single-hop broadcast with no peer relay, and a simplified pseudonym lifecycle, as detailed in Chapter 2. This thesis does not extend the codebase toward full standards compliance; where a simplification materially affects an attack’s external validity, this is flagged explicitly in Chapter 3 rather than corrected.

- **Switching simulation stack.** All experimental work is conducted on Artery/SUMO/OMNeT++, the stack used by both prior theses. No comparison to, or migration toward, an alternative simulator (e.g. Veins, ns-3) is undertaken. This also fixes the radio access technology at ITS-G5, as implemented by Artery’s underlying Vanetza library; comparison against the competing C-V2X radio access technology is out of scope for the same reason (Chapter 2, Section 2.1.1) — the revocation schemes under study operate above this layer and are not expected to depend on which one carries the traffic.

1.6 Thesis Structure

The remainder of this thesis follows the Plan–Do–Check–Act structure set out in the thesis template.

Chapter 2, State of the Art, surveys V2X communication and PKI fundamentals, describes the three revocation schemes and their prior evaluation, and reviews the existing attack literature on V2X revocation and credential management, concluding with the literature gap this thesis addresses.

Chapter 3, Threat Model and Attack Selection (Plan), formalises the attacker model motivated in Section 1.1 and grounded in Chapter 2, defines the six-criteria methodology used to select attacks, and presents the resulting attack catalog together with the candidates excluded and the rationale for excluding them.

Chapter 4, Experimentation (Do), describes the corrective code changes required before meaningful comparison is possible (most importantly, closing the trivial re-enrolment gaps confirmed by code inspection), the implementation of each selected attack, and the experimental design and parameters for each.

Chapter 6, Analysis (Check), presents the experimental results per scheme and per attack, answering RQ1 quantitatively and RQ2 architecturally.

Chapter 7, Discussion (Check), addresses RQ3: detection signatures and candidate mitigations for each attack found to degrade a scheme, and whether adopting them would change the comparative robustness ranking established in Chapter 6.

??, Conclusions, summarises the thesis’s answers to RQ1–RQ3, states its limitations, and identifies directions for future work that this thesis’s own non-goals deliberately leave open.

Chapter 2

State of the Art

This chapter surveys the technical and academic background required to situate the thesis’s contribution. Section 2.1 introduces the V2X communication and public-key infrastructure (PKI) context in which certificate revocation operates. Section 2.2 describes the three revocation schemes evaluated in this thesis, their design goals, and the prior evaluation performed by Dehaes [6] and Orban [15] on the same simulator this thesis extends. Section 2.3 surveys the existing attack literature relevant to V2X revocation and credential management. Section 2.4 synthesises these three bodies of work into the specific gap this thesis addresses, which is then operationalised into a threat model and attack selection methodology in Chapter 3.

2.1 V2X Communication and Security Fundamentals

2.1.1 Safety-critical messaging and the case for a PKI

Vehicle-to-Everything (V2X) communication allows vehicles, roadside infrastructure, and other road users to exchange status and hazard information over wireless links, most commonly Cooperative Awareness Messages (CAMs) and Decentralized Environmental Notification Messages (DENMs) in the European ETSI ITS-G5 stack. ITS-G5 is the European profile of the same IEEE 802.11p-based radio access technology that SAE standardises in North America as Dedicated Short-Range Communications (DSRC) [19]; the principal competing alternative, 3GPP’s Cellular V2X (C-V2X), instead carries V2X traffic over LTE/5G sidelink or cellular infrastructure [19]. The codebase this thesis extends is built on Artery’s ITS-G5/Vanetza stack, and switching radio access technology is outside this thesis’s scope (Chapter 1, Non-Goals); the revocation mechanisms studied here operate at the certificate and application layer and are, in principle, agnostic to which of these radio technologies carries the underlying messages. Because these messages directly influence safety-relevant decisions — collision warnings, intersection assistance, emergency braking — a forged or tampered message is not merely a confidentiality problem but a safety hazard: an attacker who can inject a credible false CAM can induce another vehicle’s automated systems to brake or swerve unnecessarily. Authenticity and integrity are therefore first-order requirements, and every major V2X security architecture answers them with a Public Key Infrastructure (PKI): every outgoing message is digitally signed, and every receiving station verifies the signature against a certificate chain rooted in a trusted authority.

A PKI alone, however, creates a new problem that is central to this thesis. If every vehicle used a single long-lived certificate to sign its messages, any receiver could trivially link successive messages to the same certificate and track that vehicle’s movements over time — a direct privacy violation. V2X PKI architectures therefore issue *pseudonym certificates*: short-lived, frequently-rotated certificates that authenticate individual messages without revealing a stable, linkable identity across a longer time window. This design choice is the reason certificate revocation in V2X is materially harder than in classical PKI settings such as TLS: a compromised or misbehaving vehicle does not hold one certificate to revoke,

but a stream of pseudonyms, some already issued and cached locally by the vehicle, that must all be invalidated without breaking the privacy guarantees the pseudonym scheme was built to provide.

2.1.2 The ETSI C-ITS security architecture

In Europe, the security architecture for Cooperative Intelligent Transport Systems (C-ITS) is defined by ETSI, principally in TS 102 940 [9] (overall architecture) and TS 102 941 [10] (trust and privacy management). The architecture separates two credential types, each issued by a distinct authority:

- The **Enrolment Authority (EA)** authenticates an ITS Station (ITS-S) — an On-Board Unit (OBU) in a vehicle, or a Roadside Unit (RSU) — against its canonical, long-term identity and issues a *long-term enrolment credential*. This step happens once, or infrequently, and is comparable to device provisioning in a classical PKI.
- The **Authorization Authority (AA)** accepts an authorization request backed by a valid enrolment credential and issues one or more short-lived *Authorization Tickets (ATs)* — the ETSI term for pseudonym certificates. The AT, not the enrolment credential, is what signs day-to-day V2V and V2I traffic.

A Root Certificate Authority (RCA) anchors trust for both the EA and the AA. Critically for revocation, the EA and AA are functionally and, in a full deployment, organisationally separate: the AA that issues pseudonyms need not learn the vehicle’s long-term identity, and the EA that knows the long-term identity need not see which pseudonyms were derived from it. This separation is what allows ETSI’s revocation strategy (Section 2.2.2) to work without a central authority ever compiling a list of currently-valid pseudonyms tied to real vehicles.

The North American counterpart, the Security Credential Management System (SCMS) defined around IEEE 1609.2 and detailed by Brecht et al. [5], follows a broadly analogous enrolment/authorization split but enforces an even stricter separation of duties — for instance splitting the linkage-authority role in two so that no single organisation can deanonymise a vehicle unilaterally — and, importantly for this thesis, adopts a fundamentally different revocation strategy (Section 2.2.1). ETSI and SCMS are the two architectures referenced throughout the revocation literature surveyed below; this thesis’s codebase is modelled on the ETSI entity roles but, consistent with the simplifications documented by Dehaes and Orban, collapses EA, AA, and the revocation authority (RA) role into a single simulated authority per scheme, a simplification this chapter treats as given and does not re-litigate.

2.1.3 Revocation as a security primitive

Scopelliti et al. [16] draw a distinction that is useful for scoping this thesis precisely: *identity-based revocation* excludes a vehicle from the system based on its canonical, long-term identity — because of a policy violation, a leaked private key, or a vendor-wide vulnerability disclosure affecting a whole product line — whereas *message-based revocation* excludes a vehicle based on the content of its peer-to-peer traffic, as adjudicated by a misbehaviour detection system (MDS) that reports suspect messages to the revocation authority. Both trigger the same downstream mechanism: once a decision to revoke is

made, the vehicle’s long-term credential is blacklisted (denying future re-enrolment) and its currently-valid pseudonyms must be made unusable across the network before other vehicles will accept messages signed with them.

This thesis is concerned exclusively with what happens *after* a revocation decision has already been made — how the decision propagates, how quickly it takes effect, and how each scheme’s propagation mechanism can be attacked. It does not implement or evaluate a misbehaviour detection system; the trigger for revocation is treated as an already-authenticated, exogenous event, consistent with the scope boundary discussed in Section 2.3.5 and formalised in the threat model of Chapter 3.

2.2 Certificate Revocation Schemes

Three revocation strategies dominate the V2X literature and are implemented in the code-base this thesis extends: *active*, *passive*, and *self*-revocation. Table 2.1 summarises their enforcement mechanism and the party responsible for enforcing it; the following subsections describe each in turn.

Table 2.1: Enforcement mechanism and responsible party for the three revocation schemes under evaluation.

Scheme	Enforcement mechanism	Enforced by	Distribution
Active	Certificate Revocation List (CRL) check	Every receiving vehicle	Push, broadcast
Passive	Pseudonym expiry + enrolment denial	RA/AA, at re-issuance time	Pull, no broadcast
Self	Peer-cooperative credential deletion	The revoked vehicle’s own TC	Push, broadcast (HB)

2.2.1 Active revocation

Active revocation, the strategy adopted by SCMS [5], is the most direct extension of the classical X.509 CRL model to V2X. When the RA decides to revoke a pseudonym, it adds an identifier for that pseudonym — a hash of the public key, in the schemes surveyed here, rather than the pseudonym itself, to limit unnecessary exposure — to a Certificate Revocation List, which is then distributed to all vehicles, typically via RSU broadcast. Every vehicle checks every incoming V2V message’s signing certificate against its locally cached CRL before accepting the message.

The scheme’s principal advantage is that revocation, once the CRL update is received, is immediate and requires no cooperation from the revoked vehicle. Its principal weakness, discussed at length by Yoshizawa et al. [19] and reflected in both prior theses on this code-base [6, 15], is CRL distribution itself: as the number of revoked pseudonyms grows, CRL size grows with it, and delivering an up-to-date CRL to every vehicle in a timely manner — especially in dense traffic or under constrained bandwidth — is a genuinely hard distribution problem. Yoshizawa et al. state plainly that a CRL update is of little value if it does not reach the vehicles that need it when they need it, which is precisely the property this thesis’s attack catalog (CRL flooding, selective forwarding, coverage-gap; Chapter 3) targets directly. Khodaei and Papadimitratos [14] address this same distribution bottleneck with a vehicle-centric redesign of CRL distribution that filters pieces by each vehicle’s trip and region and validates them for free via a Bloom-filter fingerprint. Their own baseline configuration, against which they benchmark this redesign, also uses RSU broadcast and car-to-car epidemic relay, but distributes the full, unfiltered, individually-signed CRL without this filtering or free validation. The active-revocation implementation evaluated in this

thesis has no peer relay at all — a single-RSU, single-hop broadcast topology confirmed by code inspection — making it more constrained than either of their two configurations, a point returned to in Section 2.3.2.

2.2.2 Passive revocation

Passive revocation, ETSI’s chosen strategy [10], avoids distributing any revocation-specific message at all. Pseudonym certificates are issued with a short, fixed validity period; a vehicle must periodically return to the AA to obtain fresh pseudonyms. When the RA decides to revoke a vehicle, it does not touch already-issued pseudonyms — it blacklists the vehicle’s long-term enrolment credential internally and simply declines the vehicle’s next re-authorization request. ETSI TS 102 941 §6.1.4 formalises this as the canonical definition of passive revocation: enforcement happens exclusively at the reload point, not at the receiving peer, which is why neither active revocation’s CRL nor self-revocation’s heartbeat has any counterpart in the passive scheme.

The scheme’s principal advantage is architectural simplicity and reduced network overhead — no broadcast channel dedicated to revocation state exists at all, and consequently, as shown in Section 2.2.4 and argued further in Chapter 3, no broadcast-disruption attack has anything to disrupt. This structural property, not an oversight in this thesis’s attack selection, is why passive revocation appears as immune or partially resilient against several of the attacks catalogued in Chapter 3. The scheme’s well-documented weakness is the mirror image of its strength: a revoked vehicle retains full communication capability until its last valid pseudonym expires, a window bounded below by the chosen pseudonym lifetime. Shortening that lifetime reduces the window but increases re-authorization traffic proportionally, a trade-off both Dehaes and Orban discuss and that neither prior thesis nor this one attempts to resolve, since doing so would mean proposing a new scheme, which is explicitly out of scope for this thesis.

2.2.3 Self-revocation

Self-revocation departs from both preceding designs by requiring the revoked vehicle’s own hardware to cooperate in its own exclusion. The strategy was first proposed by Förster et al. as REWIRE [12]: each vehicle is assumed to contain a Trusted Component (TC) — conceptually a TPM or TEE — that independently manages the vehicle’s cryptographic material. When the RA decides to revoke a vehicle, it broadcasts an Order for Self-Revocation (OSR) containing the target pseudonym’s identifier; the target’s own TC recognises the identifier as its own and deletes its credentials, after which it can no longer produce valid signatures. Because resolution to a real-world identity is never required, the scheme preserves privacy while achieving what is, in principle, immediate revocation with no CRL distribution problem.

REWIRE’s original design depended on a second, parallel mechanism to bound the case where a vehicle simply ignores the OSR: periodic RA-broadcast keep-alive messages, whose absence for an extended period would trigger automatic self-revocation on the assumption that the vehicle had lost connectivity or been compromised. As both prior theses note, this dependence on a locally trusted, synchronised time source is fragile, and — a gap neither the original REWIRE paper nor the surrounding literature resolved — nothing in the design prevents a vehicle from selectively dropping only the OSR while continuing to process keep-alives normally, defeating revocation while appearing perfectly healthy to the

network.

Scopelliti et al. [16] redesign self-revocation around a single, unified mechanism that closes this gap: the *heartbeat* (HB). The RA periodically broadcasts an authenticated HB containing a *Pending Revocation List* (PRL) — the identifiers of all pseudonyms currently scheduled for revocation — and a timestamp t_{hb} . Vehicles relay every HB to their TC, which (i) verifies the RA’s signature over the PRL and timestamp, (ii) checks freshness, $t_{hb} \geq now - T_v$ for a configurable validity window T_v , and (iii) on success, advances its internal clock to $now := \max(now, t_{hb})$. If the vehicle’s own pseudonym identifier ever appears in a PRL it relayed and accepted, the TC deletes its credentials directly. Because the same message that carries revocation state also carries the only source of time the TC trusts, a vehicle cannot selectively suppress the PRL entry that targets it without also refusing to synchronise its clock at all — and an unsynchronised TC eventually desynchronises past T_v and is treated as unreachable. This single design decision is what formally bounds the *effective revocation time*: the authors define it as the interval between the RA’s revocation decision and the last verified message signed by the revoked pseudonym, and prove it has a deterministic upper bound under their four design objectives — guaranteed revocation (O1), no reliance on a locally trusted time source (O2), scalability (O3), and configurability for varying network conditions and security requirements (O4) — even against an attacker that drops or delays HBs, which they evaluate directly using three attacker sophistication tiers (*honest*, *smart*, *blind*), returned to in Section 2.3.3. The TC itself is assumed fully trusted and outside the attacker’s reach; only the untrusted OBU software that relays messages to it is considered compromised.

This design is the one implemented in the codebase this thesis extends. The thesis’s own code inspection (documented in Chapter 3) confirms two properties of the implementation worth flagging here because they recur throughout the attack catalog: enforcement at a *peer* vehicle operates exclusively through the same timestamp-desynchronisation logic used for self-enforcement — there is no separate check of an incoming sender’s certificate against a locally cached PRL, matching Scopelliti et al.’s own formalisation of peer-side enforcement rather than reflecting a simulator simplification — and HB signature verification in this implementation uses a key embedded in the HB message itself rather than a pre-provisioned RA certificate, a genuine simulator simplification rather than a protocol flaw, but one with direct consequences for the trusted-time attacks discussed in Section 2.3.3.

2.2.4 Prior comparative evaluation

All three schemes above were implemented side by side, in three otherwise comparable simulation scenarios, by Dehaes [6] and Orban [15] on the Artery/SUMO/OMNeT++ stack this thesis continues to use. Their combined evaluation, restricted to benign network conditions — random packet loss and delay applied uniformly at the RA and radio layers, with no adversarial vehicle behaviour — is the direct empirical predecessor of this thesis and its baseline for effective revocation time and network overhead. Their headline finding is that active and self-revocation achieve statistically similar median effective revocation times, both propagating revocation state at essentially the same rate, while passive revocation’s revocation time is, by construction, bounded below by the configured pseudonym lifetime and is therefore substantially higher across all tested parameter settings — an expected and design-intentional result rather than a deficiency, since the experiment set the pseudonym lifetime equal to the target revocation-time baseline. Neither thesis evaluates any condition in which a vehicle behaves adversarially with respect to the revocation mechanism itself;

the packet loss they model is a network condition applied uniformly to all traffic, not a strategy chosen by a rational attacker who specifically wants to evade revocation. This is the precise starting point this thesis’s own inspection of the codebase reconfirmed before selecting any attack, detailed further in the discussion of A1’s novelty (Section 3.3).

A second point of comparison is worth noting for scope reasons. Angelogianni et al. [2] independently compare a PKI-based architecture against a Direct Anonymous Attestation (DAA)-based one, including each architecture’s revocation capability, and measure signing/verification cost, end-to-end latency, pseudonym-reload efficiency, and revocation execution time across both. This is one of the few published attempts at a systematic, quantitative comparison across revocation strategies, and Orban explicitly flags it as the closest prior work to a comparative revocation framework in the literature. It differs from this thesis, and from Dehaes and Orban’s own comparison, along two axes that matter for how this thesis’s contribution should be framed: it compares architectures (PKI versus DAA), not revocation *schemes* within a single architecture, and — like Dehaes and Orban’s evaluation — it does not introduce any adversarial condition; all measurements are taken under normal operation. No source found in this literature review evaluates active, passive, and self-revocation together under a common, deliberately adversarial condition, a point developed fully in Section 2.4.

2.3 Existing Attack Literature on V2X Revocation and Credential Management

2.3.1 Attacker models and threat taxonomies

Two complementary frameworks recur throughout the literature reviewed for this thesis and are adopted directly in its own threat model (Section 3.1). Yoshizawa et al. [19] classify V2X attackers along five independent dimensions — active/passive, local/remote scope, local/global view, insider/outsider, and individual/organised — which allows a precise, composable statement of attacker capability rather than a single coarse label such as “adversary.” Separately, formal-verification work on revocation protocols, including Scopelliti et al. [16] and Whitefield et al. [18], models the network itself as a Dolev-Yao adversary: an attacker who fully controls the network — able to intercept, reorder, delay, replay, and suppress any message — but who cannot break the underlying cryptographic primitives or compromise a vehicle’s TC. This is the dominant attacker model for revocation-specific work because it isolates protocol-level weaknesses from cryptographic ones, and it is the model this thesis inherits for its own primary attacker (Section 3.1). At the standards level, ETSI TR 102 893, the C-ITS threat, vulnerability, and risk analysis [7], provides a structured threat-ID vocabulary — including categories for message suppression, replay, and denial of service against V2X infrastructure — that this thesis uses to ground several attacks in a recognised standards taxonomy rather than an ad hoc description.

2.3.2 Attacks on broadcast-based revocation distribution

Both active revocation’s CRL broadcast and self-revocation’s heartbeat broadcast share a structural dependency on reliable message delivery, which the literature identifies as an attack surface from two directions.

Selective, receiver-side suppression. Sedar et al. [17] characterise the *gray-hole* attack as a compromised node that continues to relay other traffic normally while selectively

dropping only the data packets belonging to one targeted application or service. Van der Heijden et al. [13] describe the same underlying pattern in more general terms, without using this specific label, as a routing-layer attacker that deviates from expected behaviour to cause specific messages to be lost rather than disrupting the network indiscriminately. This thesis adapts the pattern to a revocation-distribution channel: a compromised node continues to participate normally in all other V2X traffic, but selectively fails to process the specific messages — the CRL update or heartbeat — that would trigger its own exclusion, treating the revocation service itself as Sedar et al.’s targeted application. Because this class of attack targets message *processing at the receiver*, not the distribution channel itself, it applies differently to each scheme depending on where enforcement actually happens: it has direct purchase against active revocation (a vehicle simply never updates its local CRL) and self-revocation (a vehicle simply never relays HBs to its TC), but has no target at all against passive revocation, whose enforcement point is the RA’s re-authorization check, not anything the attacker’s own OBU processes.

Distribution-channel flooding and pollution. Khodaei and Papadimitratos [14] evaluate the opposite direction: rather than suppressing delivery to itself, a coalition of compromised, validly credentialed vehicles floods the CRL distribution channel with excess or fabricated CRL pieces, degrading delivery to *everyone*. Their evaluation methodology — tracking the percentage of “cognizant” vehicles (those holding an up-to-date CRL) over time as a function of attacker population share (0%, 1%, 5%, 10%, 25%, and 50% of vehicles) — is, to date, the most directly reusable experimental design found in this review for measuring a distribution-layer attack’s effect, and is adopted as a methodological template in Chapter 3. Their result that the baseline scheme degrades sharply once attackers exceed roughly 10% of the vehicle population, while their vehicle-centric scheme remains stable even at 50%, is directly relevant here — though both of their configurations use RSU broadcast and peer relay; the resilience gap comes from the vehicle-centric scheme’s trip-relevant filtering and free validation, not from relay presence. The active-revocation implementation evaluated in this thesis has no peer relay at all and no per-piece validation, making it more constrained than either of Khodaei and Papadimitratos’s configurations, a scope difference this thesis states explicitly rather than implying a stronger result than the evidence supports.

2.3.3 Attacks on self-revocation and trusted time

Self-revocation’s dependence on the HB channel for both revocation state and time synchronisation makes it the most attack-studied of the three schemes in the literature, along two distinct lines.

Direct evaluation of HB-channel attackers. Scopelliti et al. [16] evaluate their own scheme against three attacker sophistication tiers operating on HB delivery to the TC: an *honest* baseline that relays every HB as received; a *smart* attacker that relays HBs normally but silently drops any HB whose PRL contains its own pseudonym, while continuing to synchronise on all others; and a *blind* attacker, modelling a scenario where HB content is encrypted so the attacker cannot inspect the PRL before deciding whether to relay, forced instead to guess — the best strategy the authors found was relaying exactly one HB per validity window T_v , just enough to avoid triggering desynchronisation while minimising exposure to revocation content. Even against the smart attacker — the strongest tier they test — effective revocation is bounded, because the honest peers’ clocks keep advancing past $t_{rev} + T_v$ regardless of what the attacker relays. This result is important to this

thesis for two reasons: it is the direct theoretical precedent for the PRL-suppression attack in this thesis’s own catalog (Chapter 3), and it is evaluated only within the self-revocation scheme, in isolation — it says nothing about how an equivalent-capability attacker would fare against active or passive revocation running the same underlying network topology, which is precisely the comparative question this thesis asks.

Formally identified protocol flaws. Whitefield et al. [18] apply the Tamarin prover to REWIRE-family self-revocation protocols and identify two distinct flaw classes. In the “plain” protocol variant, a vehicle that changes its pseudonym before receiving an OSR referencing its old pseudonym silently ignores the message — a functional correctness gap rather than an active attack, since it requires no adversarial behaviour, only unlucky timing. A gap Förster et al.’s own original REWIRE paper had already flagged, which Whitefield et al.’s formal analysis independently reconfirmed. In the “R-token” variant, they identify a genuine authentication flaw: the confirmation message a vehicle sends back to the RA to acknowledge self-revocation can be forged by any adversary who observes the broadcast OSR, because the RA has no pre-established key against which to verify that the confirmation actually originated from the targeted vehicle. The authors’ proposed fix, the “O-token” scheme, binds a dedicated, RA-verifiable key pair to each pseudonym specifically for revocation confirmation. Neither flaw transfers directly to the HB/PRL design evaluated in this thesis, which has no OSR/confirmation exchange at all — but the underlying *flaw pattern*, a message accepted as authoritative because the verification key is not anchored to an independently trusted, pre-distributed credential, is structurally identical to the self-referential HB signature verification this thesis’s own code inspection identified in the simulator (Section 2.2.3). This thesis treats Whitefield et al. as formal precedent establishing that this class of flaw is a recognised risk in this protocol family, not evidence that the specific simulator behaviour is itself a formally proven flaw — that claim would require applying the same formal method to this codebase, which is out of scope here.

Trusted time as an open problem. A third line of work, concerns whether a TC can be assumed to have access to reliable time at all. Alder, Scopelliti, Van Bulck, and Mühlberg [1] characterise temporal guarantees in Trusted Execution Environments as an unsolved problem: under a realistic attacker model, a TEE cannot, in general, be assumed to have access to a trustworthy local clock, which is precisely the constraint self-revocation’s HB-based synchronisation (objective O2, Section 2.2.3) was designed around rather than assumed away. That this is a genuine, actively researched problem rather than a contrived simulator artefact is confirmed by the wider trusted-time literature: Bettinger et al. [4] document that a prior, purpose-built distributed TEE time protocol (Triad) was vulnerable to exactly this class of attack — a single compromised node manipulating its own perceived time by delaying protocol messages to a remote time authority, an effect that could then cascade to honest peers — and design their own protocol, TriHaRd, specifically to close this gap, showing empirically that it detects and rejects this delay-based manipulation where the prior protocol could not. This is evidence that the underlying attack class is real and non-trivial to defend against, not that it remains open in the current state of the art. Anwar and Srivastava [3] complement this with a platform-level view, formalising three categories of timing attack against trusted execution environments: OS-level scheduling attacks, OS-level delay attacks on time packets, and network-level attacks against time synchronisation traffic. Taken together, this line of work establishes that HB delivery-delay — an attack requiring no forgery, no RA compromise, and no violation of any cryptographic

assumption — is not a contrived simulator artefact but the direct operationalisation of an open research problem in trusted computing, which is the framing this thesis adopts for its own trusted-time attack in Chapter 3.

2.3.4 Attacks on enrolment and credential re-issuance

A revocation scheme that correctly invalidates a vehicle’s current pseudonyms but allows the same vehicle to trivially re-obtain new ones has not actually revoked anything. Scopelliti et al.’s Requirement 6 [16] states this explicitly: only non-revoked vehicles may successfully re-enrol. ETSI TS 102 941 §6.2.3.2 [10] operationalises this requirement cryptographically, mandating that a re-enrolment or re-authorization request be authenticated by proof of possession of the vehicle’s current, valid credential — not merely a self-asserted identifier — which the associated Certificate Policy for European C-ITS [11] reinforces by treating re-keying as subject to the same authentication requirements as initial issuance. This is a narrow but consequential requirement: it is the only point in the surveyed literature where active and self-revocation’s push-based enforcement and passive revocation’s pull-based enforcement converge on an identical precondition. If it fails, it fails identically for all three schemes, regardless of how well each scheme otherwise propagates revocation state — a requirement this thesis’s own code inspection confirms the inherited codebase does not meet (Chapter 3).

2.3.5 Misbehaviour detection and the boundary of this thesis’s scope

A closely related family of attacks — Sybil attacks and false-accusation or “bad-mouthing” attacks against a misbehaviour detection system (MDS) — recurs throughout the broader V2X security literature and is well represented in the sources reviewed for this thesis. These attacks target the *decision* to revoke: an attacker either creates multiple false identities to disproportionately influence a reputation or voting mechanism, or submits fabricated misbehaviour reports to frame an honest vehicle for revocation. Yoshizawa et al. [19] identify MDS design, and in particular plausibility validation for misbehaviour reports, as an open problem largely left to individual implementations by the ETSI standards, resulting in inconsistent coverage across deployments; ETSI TR 103 460 [8] explicitly flags Sybil-capable message-based attacks against detection systems as an unresolved challenge rather than a solved requirement for current releases.

This thesis does not implement an MDS and therefore excludes Sybil and false-accusation attacks from its experimental scope, a decision justified in full in Chapter 3. The justification is stated here because it follows directly from Section 2.1.3: this thesis studies what happens after a revocation decision is made, and an MDS is a prerequisite for message-based revocation decisions to exist in the first place. Building even a minimal detection and reporting pipeline — new message types, vehicle-side plausibility logic, RA-side report processing — would constitute a substantial, independent research contribution in its own right, and is explicitly listed among this thesis’s non-goals. Readers should therefore not expect this chapter’s silence on Sybil attacks in Chapter 3 to reflect an oversight; it reflects a scope boundary stated openly and grounded in the same standards gap Yoshizawa et al. identify.

2.4 The Gap This Thesis Addresses

Three distinct bodies of prior work bear directly on this thesis, and none of them, individually or combined, answers the question this thesis asks.

First, the only existing *comparative* evaluation of active, passive, and self-revocation on a common simulation stack — Dehaes and Orban’s prior work on this exact Artery codebase [6, 15] — is restricted entirely to benign network conditions. Their packet loss and delay model is applied uniformly, representing ordinary network unreliability, not a rational actor choosing when and what to drop in order to specifically evade revocation. Their own code, confirmed by this thesis’s independent inspection (Section 3.3), contains no vehicle-side adversarial behaviour of any kind.

Second, the only body of work that evaluates a V2X revocation scheme under a genuinely adversarial, revocation-aware attacker — Scopelliti et al.’s own evaluation of their self-revocation design [16] with honest/smart/blind HB attackers — is confined to a single scheme, evaluated in isolation. It establishes that self-revocation is robust against a specific attacker capability, but offers no basis for comparison, because active and passive revocation were never subjected to an attacker of equivalent capability, on the same topology, in the same study. Whether active revocation’s CRL push or passive revocation’s expiry-based pull would degrade more, less, or differently under a comparably sophisticated attacker is a question the source literature does not answer, because it was never asked of more than one scheme at a time.

Third, the closest attempt at a genuinely cross-scheme, quantitative comparison — Angelogianni et al.’s evaluation of PKI versus DAA-based architectures [2] — compares architectural families under normal operating conditions, not revocation schemes under attack, and Orban’s own assessment of that work notes the absence of any open, reproducible framework that would let later researchers extend the comparison to additional schemes or adversarial conditions. This thesis’s own contribution can be read, in part, as beginning to answer that call, restricted specifically to revocation and to a deliberately adversarial setting.

What remains unaddressed, across all three bodies of work, is a single empirical question: under a deliberate, scheme-aware attacker — one who specifically understands and targets each scheme’s enforcement mechanism, rather than a generic source of packet loss — which of active, passive, and self-revocation degrades least, and along which axis (revocation time, message overhead, false-negative exposure window)? Answering this requires three things no single prior source provides together: (i) a common simulation environment holding topology, traffic model, and metrics constant across all three schemes, which Dehaes and Orban’s codebase already provides; (ii) a set of attacks chosen specifically because they exploit each scheme’s actual enforcement logic rather than generic network conditions, grounded in the broadcast-suppression, trusted-time, and enrolment-bypass literature surveyed in Section 2.3; and (iii) a fair, consistent basis for comparing the three schemes’ resulting robustness, not merely re-confirming that each scheme resists the specific attacker it was individually designed and evaluated against. Establishing (ii) and (iii) is precisely the task of Chapter 3, which formalises the attacker model this chapter has grounded in the literature and derives, from it and from direct inspection of the codebase, the specific attack catalog this thesis implements and evaluates.

One honest limitation of this framing should be stated before moving on, consistent with this thesis’s non-goals. The codebase evaluated here — like Dehaes and Orban’s

before it — makes deliberate simplifications relative to a full ETSI-compliant deployment: a single RSU, single-hop broadcast with no peer relay, and a simplified pseudonym lifecycle. The gap identified above is therefore a gap in *this simulator's* comparative coverage, not a claim that no organisation has ever tested these schemes against attackers on a production-grade C-ITS deployment. Where this distinction matters for a specific attack's external validity, it is flagged explicitly in Chapter 3 rather than left implicit.

Chapter 3

Threat Model and Attack Selection

3.1 Threat Model

Defining a precise attacker model is a prerequisite for any adversarial evaluation: without bounding what the attacker can observe, control, and achieve, the results of individual attack experiments cannot be meaningfully compared or generalised. This thesis adopts the five-dimensional attacker taxonomy proposed by Yoshizawa et al. [19], a framework which characterises attackers along the following independent dimensions: *active* versus *passive* (does the attacker interact, or only observe?); *local* versus *remote* scope (is the attacker bound by radio range?); *local* versus *global* view (does the attacker have partial or complete network visibility?); *insider* versus *outsider* (does the attacker hold valid credentials?); and *individual* versus *organised* (is the attacker a single motivated actor or a state-sponsored adversary?).

The primary attacker modelled throughout this thesis is *active, local-scope, insider, individual*: a single vehicle holding valid credentials that participates in the V2X network while behaving adversarially. This corresponds to the compromised vehicle modelled by Scopelliti et al. [16]: an attacker with root access to their own vehicle’s software stack who wishes to evade or delay revocation of their pseudonym certificates for as long as possible. Two supplementary attacker tiers are considered where the attack mechanism demands it. A *network attacker* — an active, local-scope entity controlling message delivery but holding no valid credentials and requiring no forgery capability — is applicable to delivery-delay attacks on the self-revocation heartbeat channel. A *compromised infrastructure* attacker, representing a malicious Revocation Authority, is considered exclusively for the secondary variant of the trusted-time manipulation attack and only after the cryptographic correction described in ?? is applied. Nation-state-level adversaries, organised multi-vehicle collusion, and attacks requiring simultaneous compromise of multiple infrastructure entities are explicitly out of scope.

In all cases, the attacker is modelled as a Dolev-Yao adversary at the network layer [16]: they may intercept, delay, replay, and suppress messages, but they cannot break established cryptographic primitives. The Trusted Component (TC) of honest vehicles is assumed to behave correctly. The Revocation Authority is treated as a trusted entity in all attacks except those that explicitly target it.

3.2 Attack Selection Methodology

3.2.1 Selection Criteria

The goal of this thesis is a comparative security benchmark: a set of experiments that, taken together, characterise how each revocation scheme degrades under adversarial pressure and permit a robustness ranking across the three schemes. This goal implies criteria that are distinct from a general novelty or coverage criterion. Six criteria are applied to each candidate attack.

Discriminative power is the primary criterion: does the attack produce measurably

different outcomes across at least two of the three schemes? An attack that degrades all three schemes identically provides no comparative basis and is assigned Low discriminative power. **Scheme specificity** captures whether the attack targets scheme-specific protocol logic (*Specific*) or the shared network infrastructure (*Shared*); scheme-specific attacks are analytically richer because they reveal properties of the protocol design rather than the deployment topology. **Simulation fidelity** assesses how accurately the simulated attack represents the real threat: direct C++ implementation of the attack mechanism in service-layer code yields High fidelity, while modelling the attack’s consequence at a higher level of abstraction — for example, an elevated packet loss rate as a proxy for physical channel jamming — yields Medium fidelity. The distinction must be stated explicitly in the methodology to bound the validity of the results. **Implementation cost** reflects the engineering effort required, ranging from a configuration-file change (Cfg) through extending an existing service class (Ext) to building a new C++ module and OMNeT++ network descriptor file (New). **Measurability** assesses whether the attack’s effect can be quantified by a metric that is meaningful and computable within the simulation framework; the primary metric set inherited from Dehaes and Orban [6, 15] consists of effective revocation time and message overhead, extended in this thesis by the proportion of vehicles correctly revoked within a time bound and the false-negative rate. **Threat model coherence** assesses whether the required attacker capability is realistic, well-defined, and grounded in the literature or in formal standards threat analyses.

An attack is included if it scores at least Medium on Discriminative power, Measurability, and Threat model coherence simultaneously. Attacks failing any one of these three threshold criteria are excluded regardless of their scores on the remaining criteria.

3.2.2 Candidate Evaluation

Eleven attack candidates were identified through a systematic review of the V2X security literature and through protocol analysis of the existing codebase [7, 8, 13, 14, 16, 18, 19]. Each candidate was evaluated against the six selection criteria. Table 3.1 summarises the scoring and verdicts. The notation used is as follows: H = High, M = Medium, L = Low; Sp. = scheme-Specific, Sh. = Shared infrastructure; Cfg = configuration-only change, Ext = extend existing class, New = new C++ module; ✓ = Included, ~ = Conditional (pending supervisor confirmation), × = Excluded.

Table 3.1: Evaluation of attack candidates against the six selection criteria. Threshold criteria (Discriminative power, Measurability, Threat model coherence) are highlighted in bold column headers.

ID	Attack vector	Discrim.	Spec.	Fidel.	Cost	Meas.	Coher.	Verdict
A1	Selective forwarding (gray-hole)	H	Sh.	H	Ext	H	H	✓
COV	Coverage gap scenario	H	Sh.	H	Cfg	H	H	✓
A5	CRL flooding	H	Sp.	H	New	H	H	✓
A6	Re-enrolment bypass	M	Sp.	H	Ext	H	H	✓
A7	Targeted HB/PRL suppression	H	Sp.	H	Ext	H	H	✓
A8	Trusted-time manipulation (delivery delay)	H	Sp.	H	Ext	H	H	✓
A2	Jamming / channel DoS	M	Sh.	M	Ext	M	M	~
A9	HB replay	H	Sp.	H	Ext	H	H	~
A3	Compromised RSU	L	Sh.	H	Ext	L	M	×
A10	Sybil / false accusation (MDS required)	M	Sh.	L	New	L	H	×
A4	GNSS / position spoofing	L	Sh.	L	Cfg	L	L	×

3.3 Included Attack Catalog

The following sections describe each included attack, its target scheme or schemes, its grounding in the literature, its planned implementation in the Artery simulator, and the metrics by which its effect will be measured. The two conditional attacks (A2 and A9) are included provisionally, subject to supervisor confirmation of attacker scope and implementation timeline respectively.

COV — Coverage Gap Scenario

This experiment varies the RSU’s radio transmission range as a single configuration parameter and measures the proportion of vehicles correctly revoked within a fixed simulation window. It probes a consequence of the single-RSU, single-hop broadcast topology already noted in Section 2.3.2 as a scope simplification relative to a multi-relay deployment, rather than a vulnerability newly discovered here: all three schemes distribute revocation information exclusively via Single-Hop Broadcast (SHB), meaning any vehicle outside the RSU’s radio range never receives a CRL update, a heartbeat message, or an enrolment denial. The experiment is strongly differentiating because passive revocation is partially resilient — existing pseudonym certificates continue to expire locally regardless of coverage, and re-enrolment is denied at the RA boundary — while active and self-revocation both fail proportionally to the size of the coverage gap. No code change is required; the experiment is driven by the `communicationRange` transmitter parameter in the scenario `.ini` file. A multi-RSU extension, adding a second stationary RSU via `numRoadSideUnits = 2` in the same file, enables a further investigation of how uncoordinated RA deployment affects revocation completeness. The primary metrics are the percentage of vehicles correctly revoked within time T and the effective revocation time, each plotted against the RSU range parameter.

A1 — Selective Forwarding (Gray-hole, Receiver-side)

A compromised vehicle receives revocation-critical messages and discards them silently instead of processing them. In the active revocation scenario, the attacker ignores incoming CRL updates and permanently retains a stale local revocation list; it continues to accept messages from a revoked peer long after other vehicles have enforced revocation. In the self-revocation scenario, the revoked vehicle ignores HB messages and prevents its own credential deletion. This attack was confirmed as a genuine new contribution through code inspection: the prior theses by Dehaes and Orban [6, 15] modelled probabilistic packet loss at the RA and radio layers to approximate benign network conditions, not as an adversarial vehicle-side suppression. The attack is grounded in the gray-hole attack characterised by Sedar et al. [17], who describe an attacker that forwards other traffic normally while selectively dropping only the data belonging to one targeted application, and in van der Heijden et al.’s [13] more general description of routing-layer attackers that selectively cause specific messages to be lost rather than disrupting traffic indiscriminately. This thesis applies that pattern to the revocation-distribution channel specifically, treating CRL updates and heartbeats as the targeted application. Passive revocation is structurally immune to this attack: its enforcement mechanism requires no broadcast delivery to receiving vehicles, relying instead on local certificate expiry and RA-side enrolment denial. The implementation adds a `MaliciousVehicleService` subclass that overrides `indicate()` to discard scheme-specific messages. The experimental axis is the proportion of attacking vehicles (1–30% of

the total fleet). For active revocation, the failure mode is permanent partial revocation. For self-revocation, peer rejection operates exclusively through timestamp desynchronisation rather than explicit PRL lookup at the receiver — matching Scopelliti et al.’s own formalisation of peer-side enforcement [16] rather than a Dehaes/Orban simplification, and confirmed by code inspection to be faithfully implemented here; accordingly, A1 converts immediate revocation into a time-bounded by the effective revocation time $T_{\text{eff}} = 2T_v$ (Scopelliti et al., Eq. 10).

A5 — CRL Flooding

A malicious vehicle with valid credentials transmits enrolment requests to the RSU at high frequency, exhausting the RA’s processing capacity and delaying legitimate CRL broadcasts. Code inspection confirmed the absence of any rate-limiting, request queuing, or origin validation in `CentralAuthService::indicate()`: every incoming packet is processed synchronously and a signed pseudonym certificate is issued unconditionally. This attack targets active revocation exclusively; passive revocation has no CRL distribution mechanism and is therefore unaffected, and the three schemes each run in a fully isolated simulation scenario, precluding any cross-service interference. The evaluation methodology follows Khodaei and Papadimitratos [14], who evaluate a structurally analogous attack on CRL distribution by measuring the percentage of cognizant vehicles over time at attacker proportions ranging from 1% to 50%. This thesis adopts the same evaluation design, restricted to the centralised, single-RSU, no-relay topology confirmed by code inspection; the scope difference is explicitly acknowledged in the experimental reporting. The attack is implemented as a `FloodingVehicleService` class that issues repeated enrolment requests in its `trigger()` cycle. The primary metrics are the percentage of vehicles with an up-to-date CRL over time and the effective revocation time, each plotted against the attacker proportion.

A6 — Re-enrolment Bypass

A vehicle whose long-term credential has been placed on the revocation blacklist attempts to re-enroll using its revoked identity or a modified identity string. Code inspection revealed two structurally distinct gaps relative to the requirements established below. For active and self-revocation, the base class `CentralAuthService::handleEnrolmentRequest()` issues credentials unconditionally — there is no blacklist check, and re-enrolment succeeds trivially, rendering effective revocation time equal to zero. For passive revocation, `PseudoAuthService` adds a check against the revocation list, but compares only the self-reported `vehicleId` string: the `EnrolmentRequest` message carries no signature and no proof of possession of the corresponding key pair, so a revoked vehicle reporting a different identity string bypasses the check without breaking any cryptographic primitive. This attack is grounded in Requirement 6 of Scopelliti et al. [16] (only non-revoked vehicles may re-enroll) and in the cryptographic binding requirement of ETSI TS 102 941 §6.2.3.2 [10], which mandates that re-enrolment be authenticated by proof of possession of the current valid enrolment credential. A prerequisite correction must be applied to the base class before this experiment can produce meaningful comparative results: adding a blacklist check to the active and self-revocation enrolment paths is necessary to ensure the comparison tests enforcement-depth differences rather than the trivial absence of any check. The metrics are the proportion of re-enrolment attempts that succeed after blacklisting and the

resulting post-revocation operational window.

A7 — Targeted HB/PRL Suppression

A vehicle targeted for self-revocation inspects the Pending Revocation List (PRL) embedded in every received HB message and silently discards any HB that contains a hash of its own pseudonym certificate, preventing the invocation of `performSelfRevocation()`. This attack is a scheme-specific, more sophisticated variant of A1: whereas A1 discards all scheme-relevant messages indiscriminately, A7 processes HBs normally whenever they do not contain the attacker’s own pseudonym hash, maintaining clock synchronisation and avoiding detection through anomalously stale timestamps until the first HB containing its hash arrives. The attack is motivated by the smart attacker tier defined in Scopelliti et al. [16] and by the authentication-gap flaw class formally identified by Whitefield et al. [18] in the REWIRE self-revocation protocol from which the scheme implemented here is derived. Code inspection confirms that `SelfRevocationVehicle::handleV2VMessage()` performs no check against a locally cached PRL: peer rejection operates exclusively through timestamp desynchronisation, whereby V2V messages timestamped more than T_v seconds behind the vehicle’s internal clock are rejected. This matches Scopelliti et al.’s own protocol formalisation [16], which derives peer-side rejection entirely from the freshness check and specifies no separate certificate lookup — the absence of explicit revocation enforcement at the receiver side is a property of the published design, not a Dehaes/Orban simplification. Consequently, A7 does not permanently prevent revocation but converts it from immediate to desynchronisation-delayed: the attacker continues communicating undetected for a window bounded by the effective revocation time $T_{\text{eff}} = 2T_v$ (Scopelliti et al., Eq. 10). The implementation overrides `handleHBMessage()` to scan the PRL before delegating to the parent class. The metric records the time from PRL entry addition to when the attacker vehicle’s messages are rejected by peers, compared against the control condition.

A8 — Trusted-Time Manipulation (HB Delivery Delay)

The self-revocation scheme synchronises each vehicle’s internal clock exclusively from the timestamp field of incoming HB messages. An attacker who controls the delivery of HB messages — delaying them without altering their content or invalidating their cryptographic signatures — prevents the vehicle’s field `mInternalClock` from advancing, which suppresses the invocation of `checkDesynchronisation()` and prevents self-revocation indefinitely. This attack is the direct simulator-level operationalisation of the TEE temporal integrity vulnerability class characterised by Alder, Scopelliti, Van Bulck, and Mühlberg [1], further explored in the TriHaRd distributed trusted-time protocol by Bettinger et al. [4] and empirically measured at the platform layer by Anwar and Srivastava [3]. The delivery-delay variant is the primary implementation target: it requires no message forgery, no RA compromise, and survives all cryptographic corrections applied in this thesis, because the attack exploits message timeliness rather than message authenticity. Code inspection confirms that `mInternalClock` is updated exclusively from incoming HB timestamps and that the freshness check enforces a lower bound only (`heartbeatTimestamp ≥ mInternalClock - T_v`) — matching Scopelliti et al.’s own freshness condition [16] (Section 2.2.3) rather than a Dehaes/Orban simplification — and therefore provides no protection against delayed delivery of authentic messages by design, not by omission. The implementation introduces a configurable processing delay in `indicate()` using OMNeT++’s

`scheduleAt()` mechanism. A secondary variant — a malicious RA subclass sending HBs with slow-advancing timestamps — is implementable after the self-referential certificate verification fix is applied and is reserved for supplementary analysis. The primary metrics are the gap between the simulator reference clock (`simTime()`) and the vehicle’s perceived time (`mInternalClock`) at the moment self-revocation should have triggered, the proportion of vehicles failing to self-revoke within the expected window T , and timestamp service availability, defined as the proportion of simulation time during which `checkDesynchronisation()` would be able to fire.

A2 — Jamming / Channel DoS (Conditional)

This experiment simulates a geographic denial-of-service by elevating the packet loss rate in a zone around the RSU, approximating the effect of physical channel jamming at Medium simulation fidelity. It is classified as conditional because the required attacker capability — an external node with no valid credentials and no V2X protocol participation — conflicts with the primary insider attacker model adopted throughout this thesis. Subject to confirmation that the outsider attacker tier is in scope, this experiment produces moderately differentiating results: active and self-revocation are both degraded by the elevated loss rate, while passive revocation is partially resilient because pseudonym certificates continue to expire locally regardless of channel conditions, and re-enrolment denial operates at the RA boundary rather than requiring broadcast delivery to vehicles. The attack is grounded in ETSI TR 102 893 threat categories V-V2 and V-R3 [7]. The simulated effect is implemented as a probabilistic drop applied in the vehicle’s `indicate()` method, scoped to a configurable geographic zone. If the outsider attacker tier is judged to be out of scope, this experiment can be reframed as a degraded-channel baseline condition rather than an adversarial attack, providing a controlled comparison with the benign-network conditions evaluated by Dehaes and Orban [6, 15].

A9 — HB Replay (Conditional)

A network attacker captures an authentic, validly-signed HB message immediately before the target vehicle’s pseudonym hash is added to the PRL, and replays it to the target within the freshness acceptance window (`mInternalClock` — T_v). The target processes a legitimate message that does not contain its own pseudonym, preventing self-revocation without any forgery, without delivery suppression, and without manipulation of timestamp values. Code inspection confirms that `HBMessage` carries no sequence number or nonce — matching the published HB format [16], which likewise specifies no such field, rather than a Dehaes/Orban simplification — and that the freshness check enforces a lower bound only: a replayed authentic message within the window is accepted without complaint by design. The attack is grounded in ETSI TR 102 893 threat category V-V6 [7]. It is classified as conditional because its measurable outcome — a $2T_v$ -bounded evasion window — overlaps with that of A7 and A8, although the underlying mechanism differs fundamentally (message substitution versus PRL-aware suppression versus delivery delay). If timeline permits, A9 is included as a standalone experiment to demonstrate that the same evasion window is reachable via three independent attack paths, each requiring a different attacker capability and exploiting a different protocol property. If timeline does not permit, A9 is merged with A7 as a variant test case.

3.4 Excluded Candidates

Three candidates were excluded from the experimental plan. The *compromised RSU attack* (A3) produces catastrophic failure across all three schemes simultaneously — the single RSU hosts the sole RA service for the entire simulation — and therefore yields no comparative signal between schemes; the attack’s practical interest as a worst-case failure mode is noted, but it does not serve the benchmark objective. The *Sybil and false-accusation attack* (A10) requires a misbehaviour detection subsystem as a prerequisite: without a detection and reporting pipeline, there is no revocation trigger to attack. Implementing even a minimal misbehaviour detector would constitute a substantial independent contribution requiring new message types, vehicle-side plausibility logic, and RA-side report processing. This scope boundary is consistent with ETSI TR 103 460 §4.2, which does not mandate misbehaviour reporting for Release 1 deployments [8], and with the finding of Yoshizawa et al. [19] that misbehaviour detection remains an open problem at the standards level. The *GNSS and position spoofing attack* (A4) was excluded because the codebase contains no geo-scoped revocation mechanism: position data is handled exclusively by the SUMO mobility layer and has no path to any CRL, PRL, or enrolment-denial logic.

Chapter 4

Implementation & Testing

4.1 Methodology

4.1.1 Testing methodology

4.1.2 Design methodology

4.1.3 Implementation methodology

4.2 Setup, Requirements, Environment, Tools & Materials

4.3 Results

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Chapter 5

Experimentation & data collection

5.1 Methodology

5.2 Setup, Requirements, Environment, Tools & Materials

5.3 Results

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Chapter 6

Experiment's output/Data Analysis

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Chapter 7

Discussion

7.1 Comparison with state of the art/related works

7.2 Lessons learned

7.3 Limitations of validity

Chapter 8

Future work

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Chapter 9

Conclusions

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